

Materials Science 2

Review of previous knowledge

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Aims of the session



- Review concepts and knowledge from previous semester
- ❖ Engineering design
- ❖ Materials Science 1: Structure and properties



Why do we need to review all this 'old' knowledge?

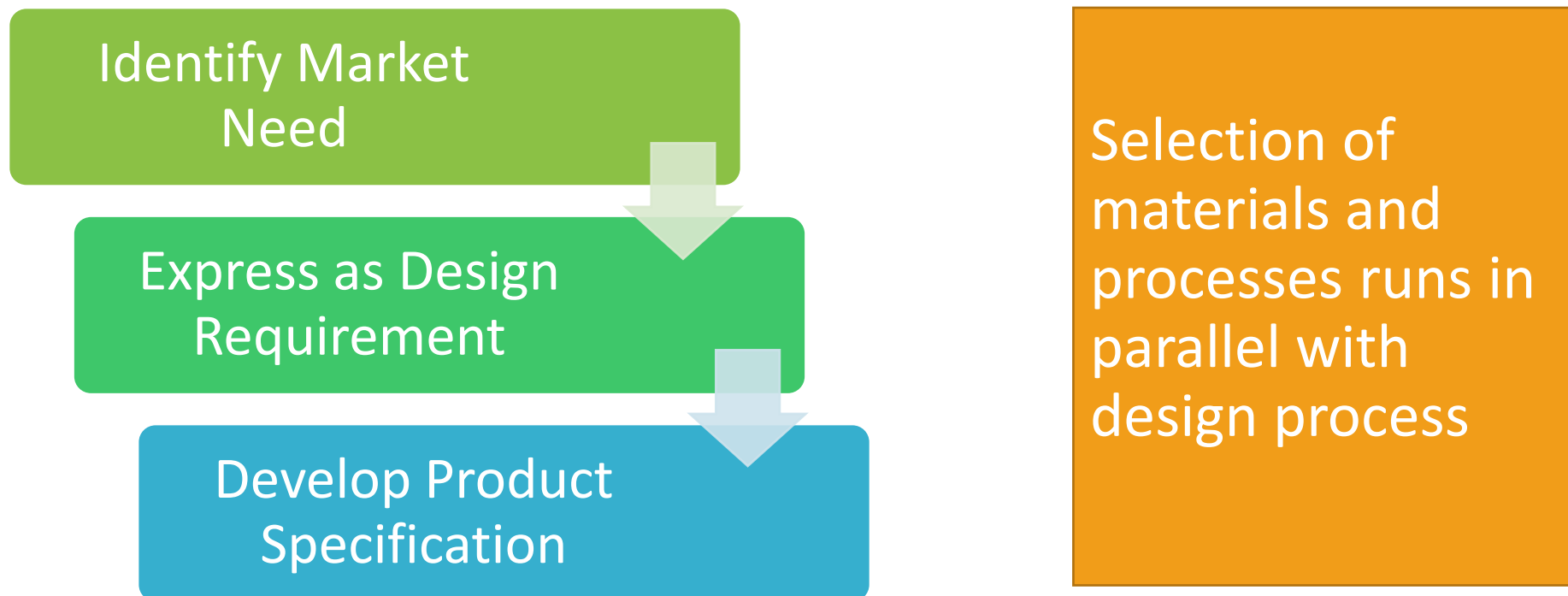
- Essential understanding of the microstructure will help you interpret concepts that we will develop later
- Changes on the microstructure occur due to phase transformations
- Metal composition and alloy structure are governed by the temperature at which the product was produced and the components that are present

Strategic thinking of material selection for design





Design-Led Approach for Selecting Materials and Processes

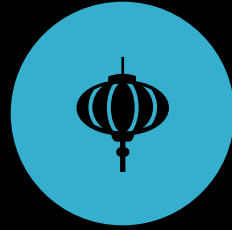




Original Design



NEW IDEA OR
WORKING
PRINCIPLE



NEW
MATERIAL
MAY
SUGGEST A
NEW
PRODUCT



NEW
PRODUCT
MAY REQUIRE
A NEW
MATERIAL



MARKET
NEED IS PUT
IN TERMS OF
A NEED
STATEMENT –
“DEVICES IS
REQUIRED TO
PERFORM
TASK X”



PRODUCT
SPECIFICATION –
DETAILED PLAN
FOR NEW
DESIGN



Material and Process Information for Design

The selection of a material begins with a catalog of all materials, which is narrowed by screening out those that fail to meet the design requirements

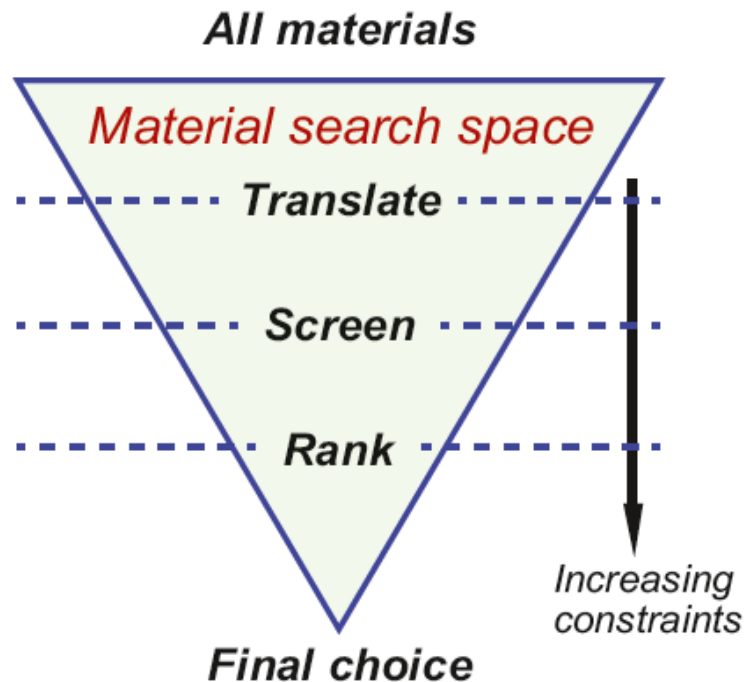
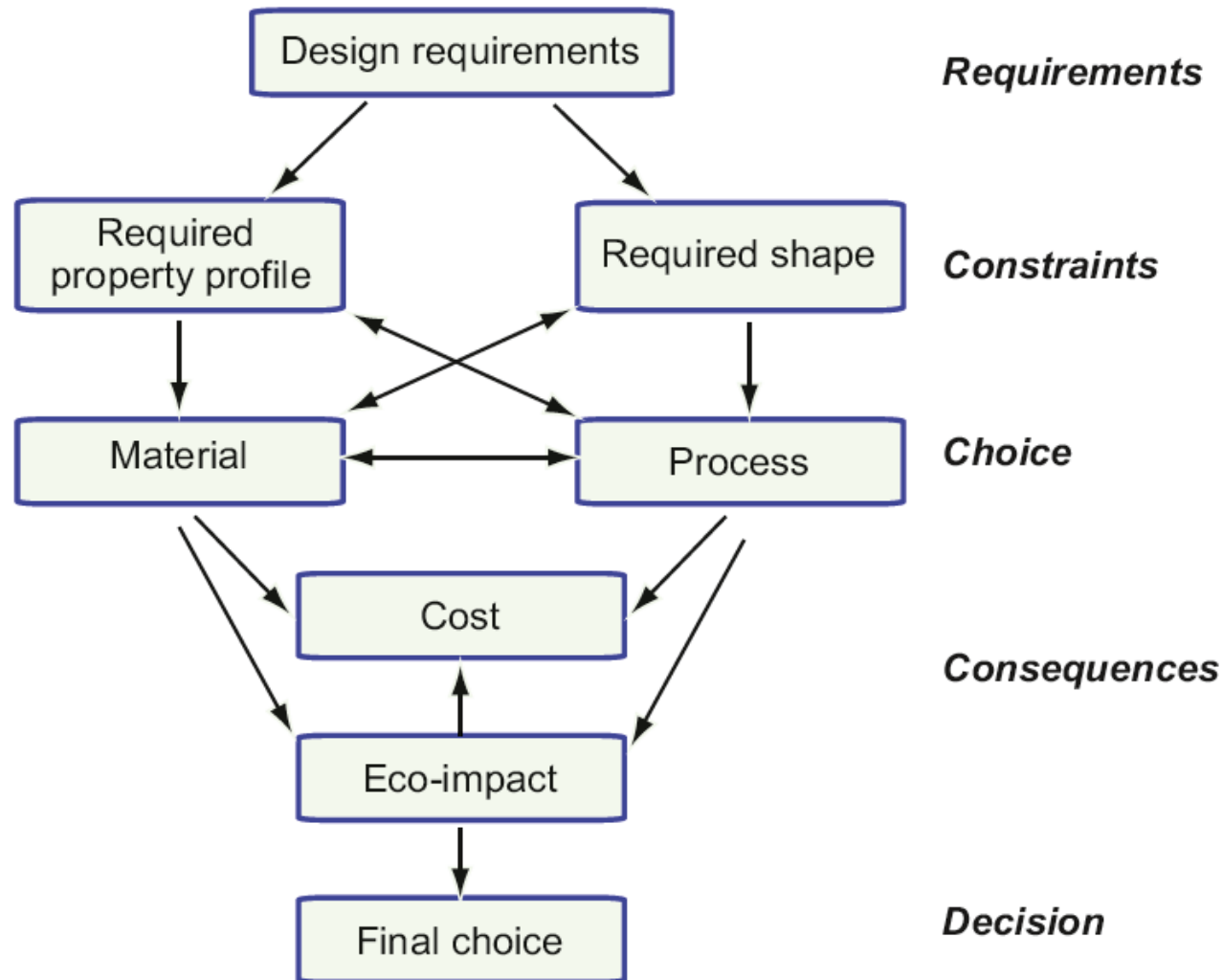


Figure 3.4

- Concept stage
 - Designer requires only approximate property values, but for widest range of material
- Embodiment stage
 - Data needed for a subset of materials, but at a higher level of detail and precision
- Detail stage
 - Even higher level of detail and precision, but for few materials



Interaction Between Design Requirements, Material, Shape, and Process

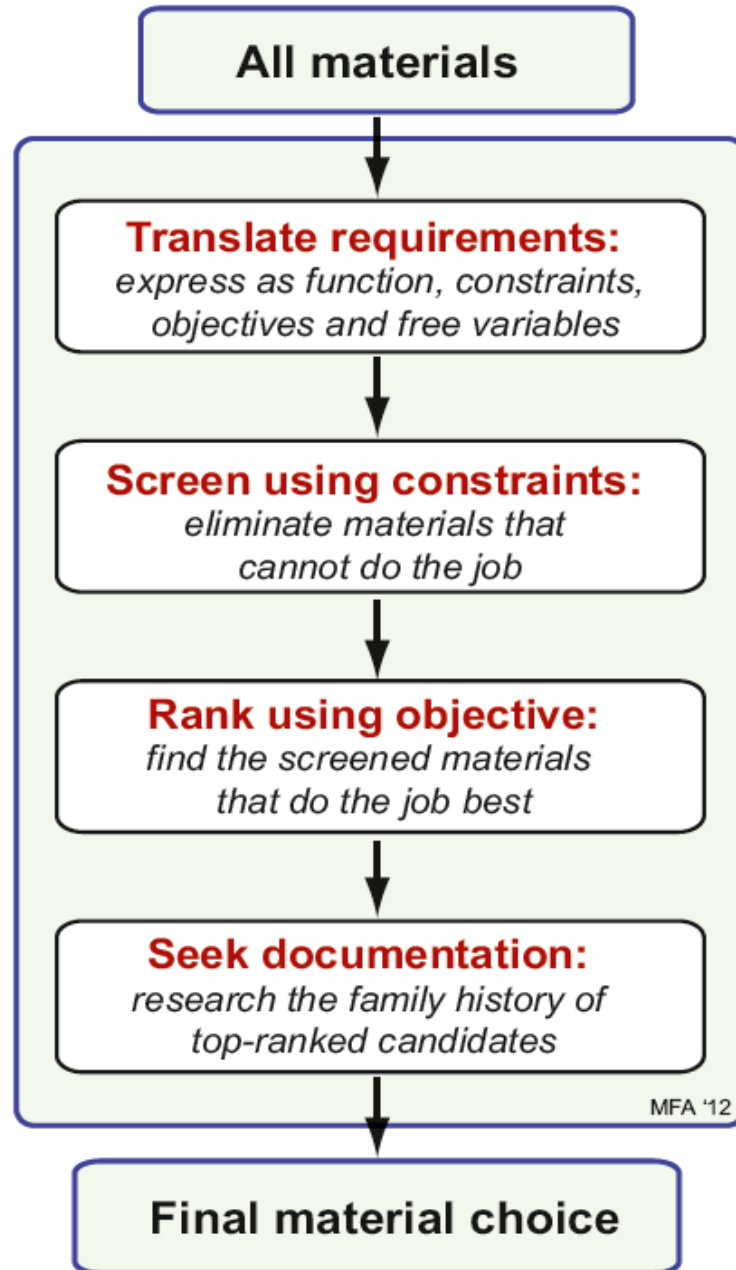


Two-Way Interactions:

Specification of shape restricts the choice of material and process, but equally the specification of process limits the materials you can use and the shape they can take



Selection Strategy



Select the best match between the attribute profiles of the materials and processes and those required by the design

4 step process

- Translation
- Screening
- Ranking
- Documentation



Translation

- **Function**
 - What does the component do?
- **Constraints**
 - What non-negotiable conditions must the material meet?
- **Objectives**
 - What aspects of performance are to be maximized or minimized?
- **Free variables**
 - What parameters of the problem is the designer free to change?

Converting the design requirements into a prescription for selecting a material

Constraints vs. Objectives

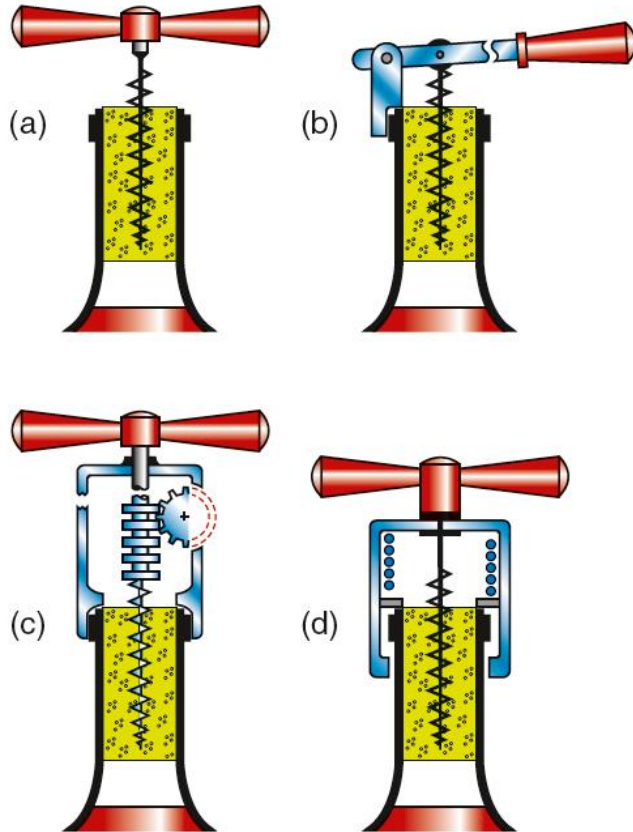
- Constraint
 - Essential condition that must be met, usually expressed as a limit on a material or process attribute
- Objective
 - Quantity for which an extreme value is sought

Common constraints*	Common objectives
Meet a target value of Stiffness Strength Fracture toughness Thermal conductivity Electrical resistivity Magnetic remanence Optical transparency Cost Mass	Minimise Cost Mass Volume Impact on the environment Heat loss Maximise Energy storage Heat flow

Example – Corkscrew lever

Requirements

- Loaded in bending when in use
- It must carry the bending moment without deflecting → high modulus, E
- It must not bend permanently → high yield strength, σ_y
- It must not snap off → adequate fracture toughness K_{Ic}
- Resistant to corrosion in wine or water
- It has a fixed length () but can have any cross-section width
- It should be of a durable material but produced cheaply



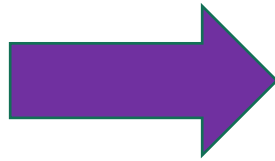
a) Direct pull; b) Levered pull;
c) geared pull; d) spring-assisted pull



Example – Cork-screw lever

Design-limiting properties

- Modulus, E
- Yield strength, σ_y
- Fracture toughness K_{1c}
- Resistance to corrosion



Translation

- Function
 - Level (beam loaded in bending)
- Constraints
 - Stiff enough
 - Strong enough
 - Some toughness
 - Resist corrosion
 - Length L (specified)
- Objectives
 - Minimise cost
- Free variable
 - Choice of material
 - Choice of cross-section area

Functional
Constraints

Worked example – Sauce-pan



This Photo by Unknown Author is licensed under [CC BY](#)

A material for an energy-efficient sauce-pan must transmit and spread the heat well, resist corrosion by foods, and withstand the mechanical and thermal loads expected in normal use. The product itself must be competitive in a crowded market.

Complete the following task

Translate these requirements into a prescription of Function, Constraints, Objectives and Free variables.



Summary



The selection of material and processing runs in parallel to product design



As the design requirements are formulated in detail, constraints emerge that must be met



Identifying the constraints, the objectives and the free variables (the process called 'translation') is the first step in selection.



To select an adequate material and a method for processing it is necessary to have a clear translation of the design requirements

Mechanical properties and microstructure of materials





Properties reflect microstructure and composition at atomic level

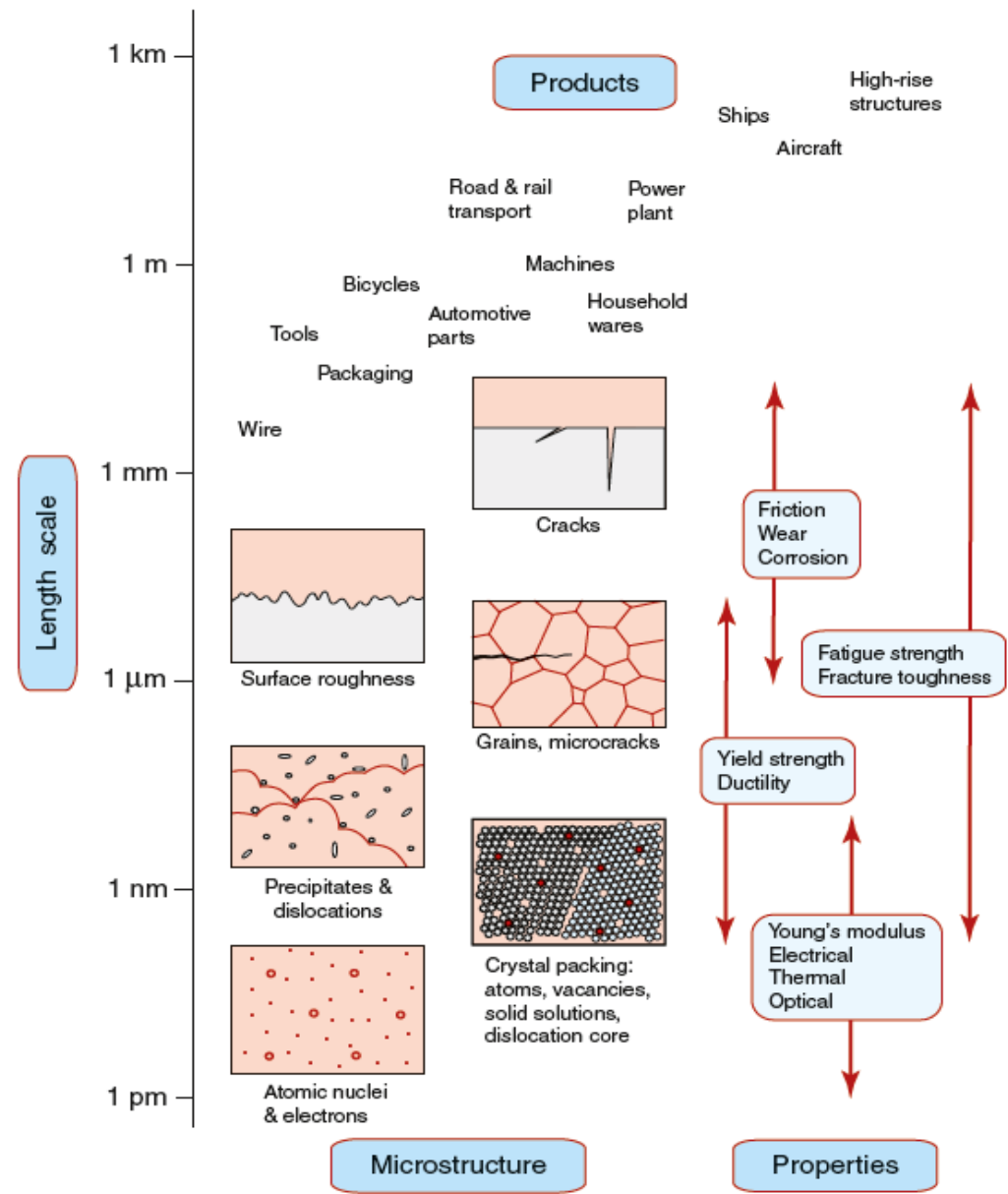
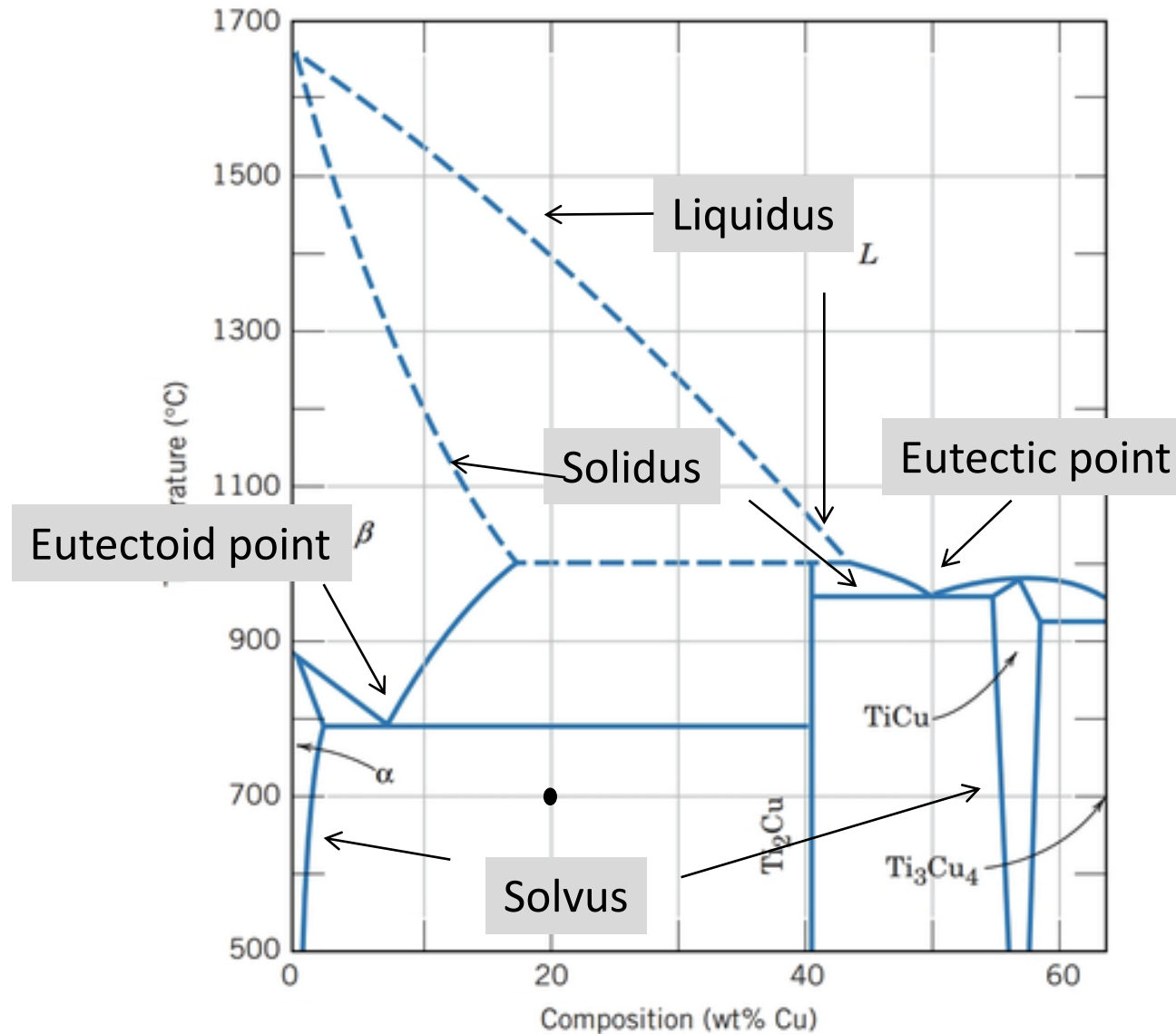


Figure 19.2 Ashby et al, 2014



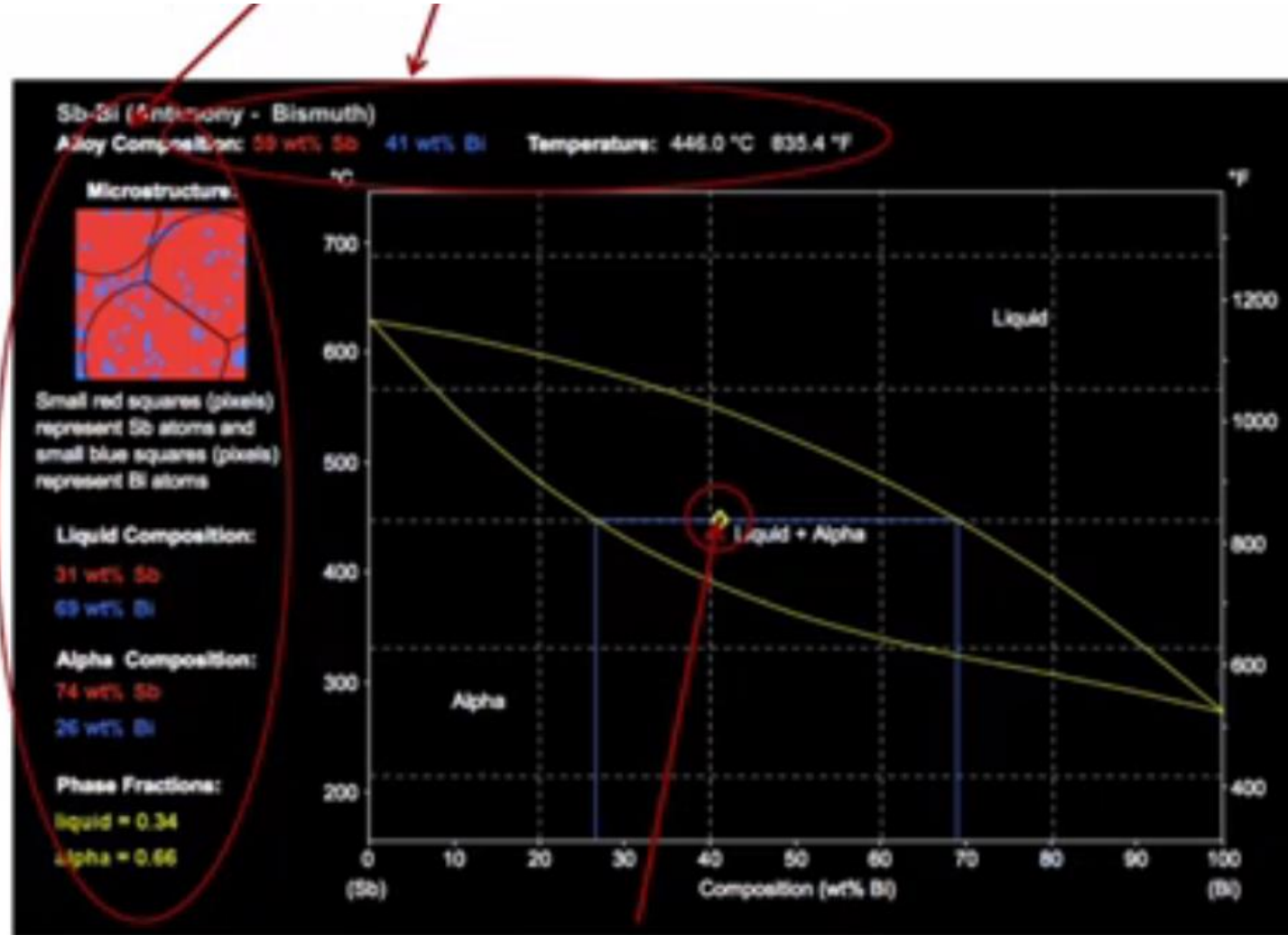
Components of a phase diagram





VMSE: Interactive Phase Diagrams

Microstructure, phase compositions, and phase fractions respond interactively



Change alloy composition



Microstructural Developments in Eutectic Systems

For alloys for which
 $C_0 < 2 \text{ wt\% Sn}$
at room temperature
there is the formation
of polycrystalline with
grains of α phase
having composition C_0

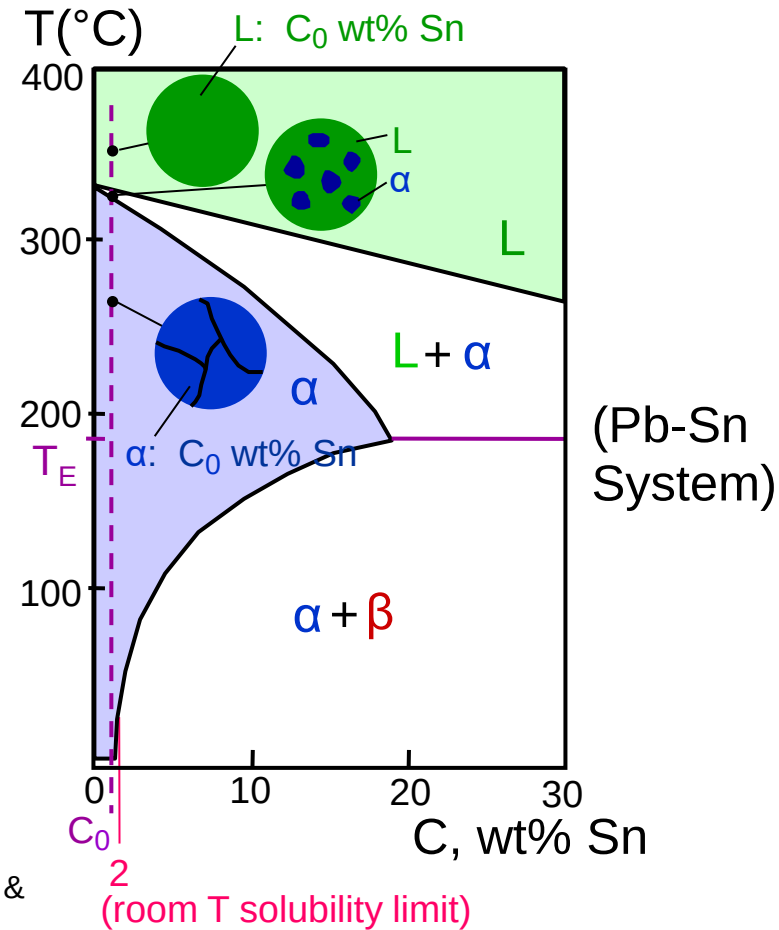
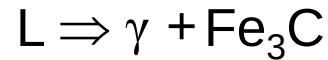


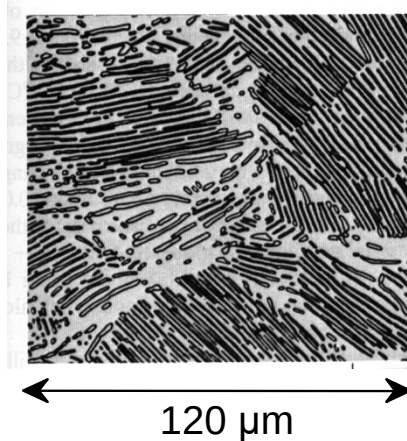
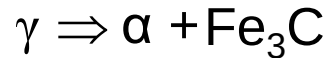
Fig. 9.11, Callister & Rethwisch 9e.

Development of microstructure in Iron-Carbon (Fe-C) alloys at eutectoid point

- Eutectic (A):



- Eutectoid (B):



Result: Pearlite = alternating layers of α and Fe_3C phases

Fig. 9.27, Callister & Rethwisch 9e.
(From Metals Handbook, Vol. 9, 9th ed.,
Metallography and Microstructures, 1985.
Reproduced by permission of ASM
International, Materials Park, OH.)

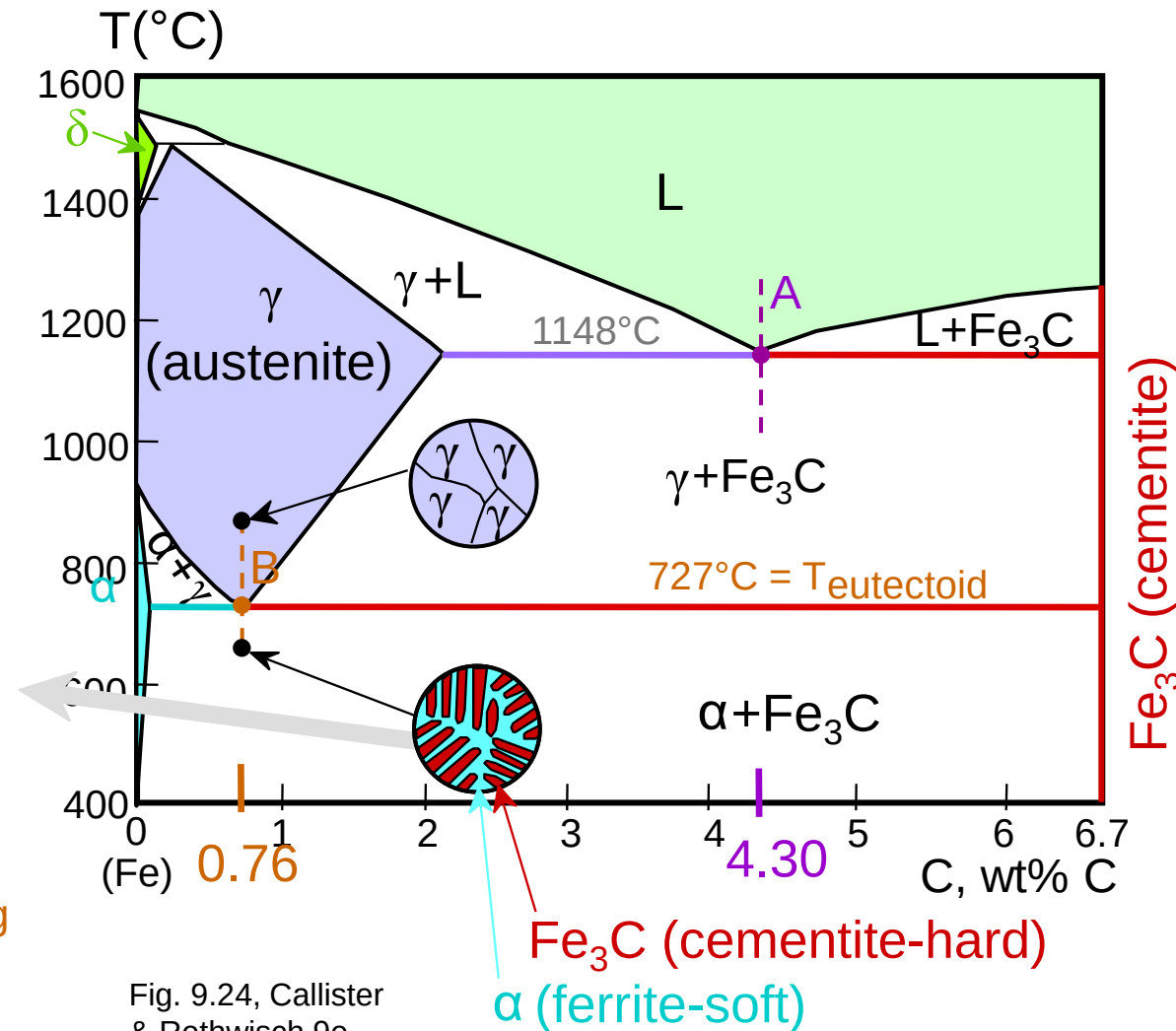


Fig. 9.24, Callister & Rethwisch 9e.

[Adapted from Binary Alloy Phase Diagrams, 2nd edition,
Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted
by permission of ASM International, Materials Park, OH.]



Austenite-to-Pearlite Isothermal Transformation

- Eutectoid composition, $C_0 = 0.76 \text{ wt\% C}$
- Begin at $T > 727^\circ\text{C}$
- Rapidly cool to 625°C
- Hold T (625°C) constant (isothermal treatment)

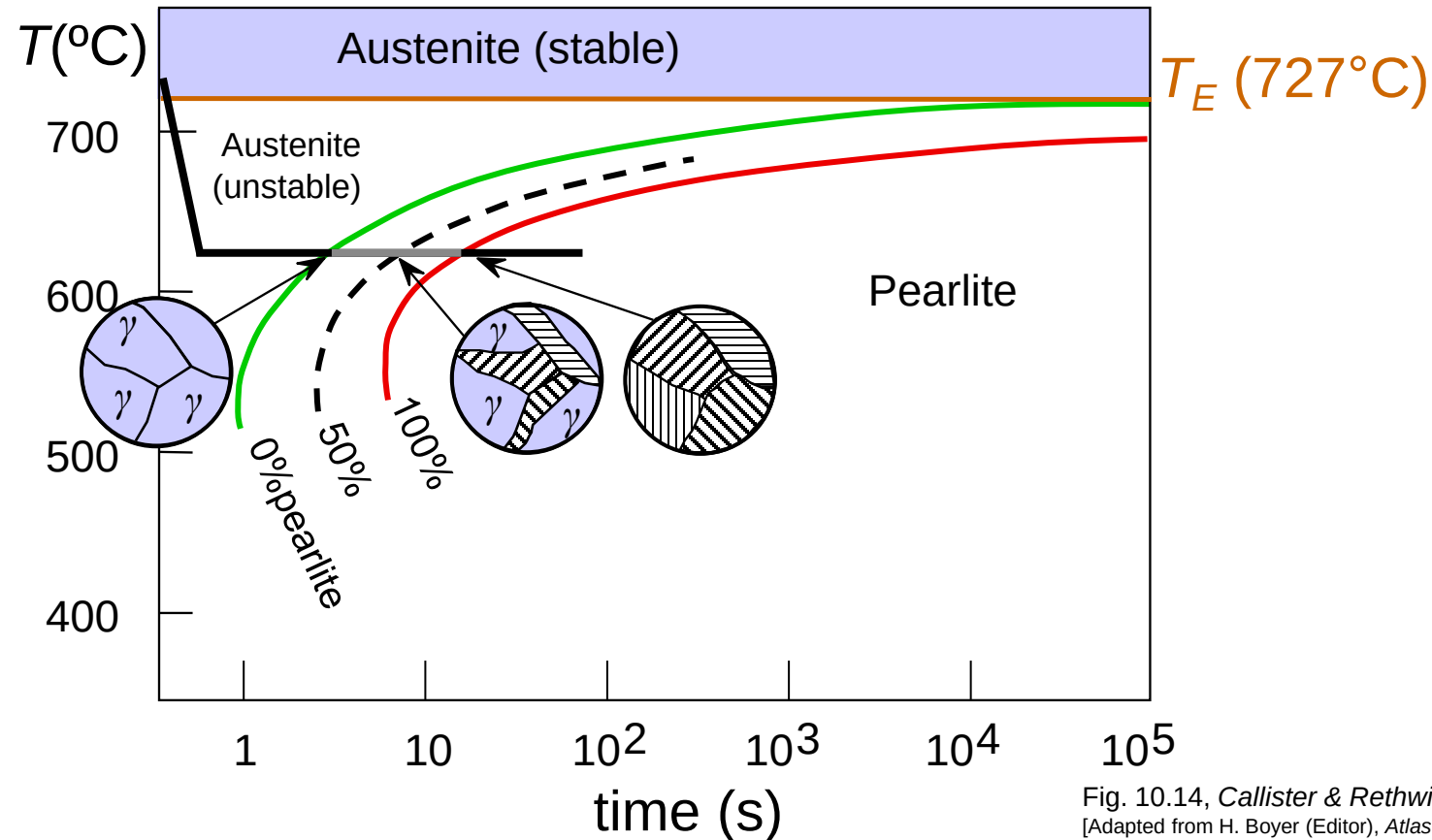
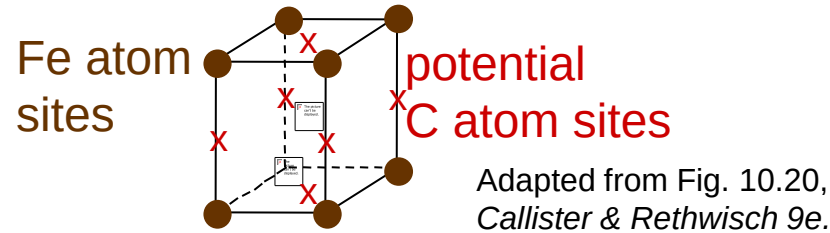


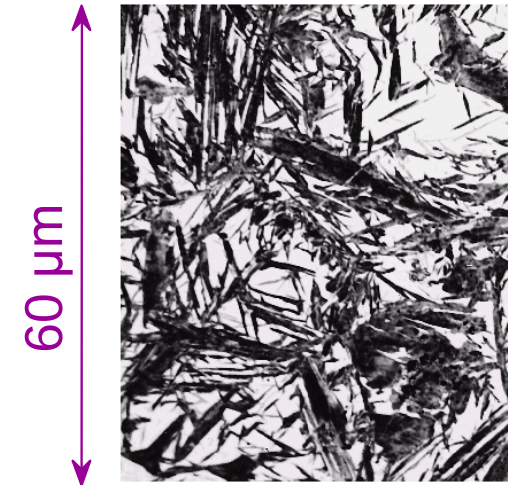
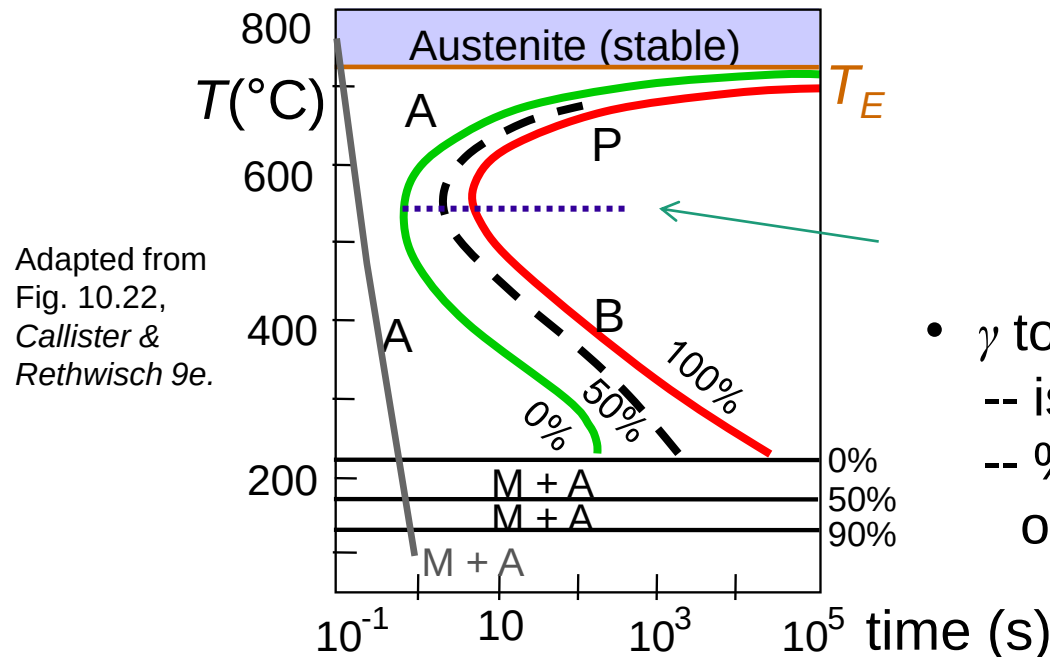
Fig. 10.14, *Callister & Rethwisch 9e*.
[Adapted from H. Boyer (Editor), *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, 1977. Reproduced by permission of ASM International, Materials Park, OH.]

Martensite: A Nonequilibrium Transformation Product

- **Martensite:**
-- γ (FCC) to Martensite (BCT)



- Isothermal Transf. Diagram



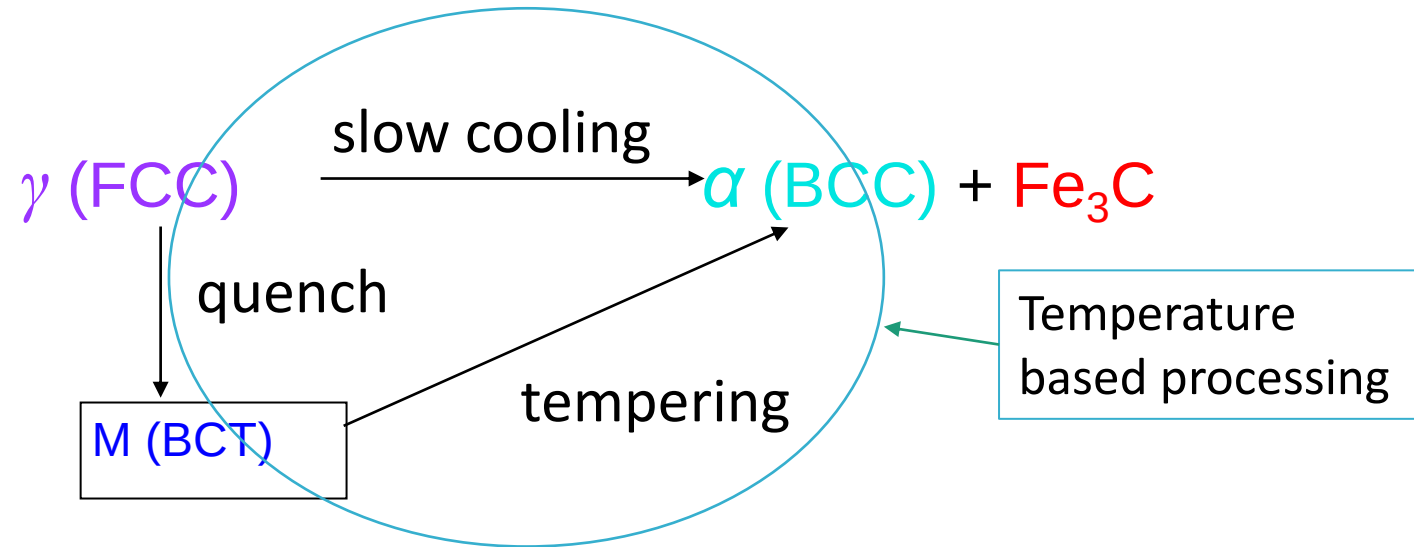
— Martensite needles
— Austenite

Fig. 10.21, Callister & Rethwisch 9e.
(Courtesy United States Steel Corporation.)

- γ to martensite (M) transformation.
-- is rapid! (diffusionless)
-- % transformation depends only on T to which rapidly cooled



Martensite Formation



Martensite (M) – single phase
– has body centred tetragonal (BCT) crystal structure

Diffusionless transformation **BCT** if $C_0 > 0.15 \text{ wt\% C}$
BCT $\longrightarrow$ few slip planes $\longrightarrow$ hard, brittle



Phase Transformations of Alloys

Effect of adding other elements

Change transition temp.

Cr, Ni, Mo, Si, Mn

retard $\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$

reaction (and formation of pearlite, bainite)

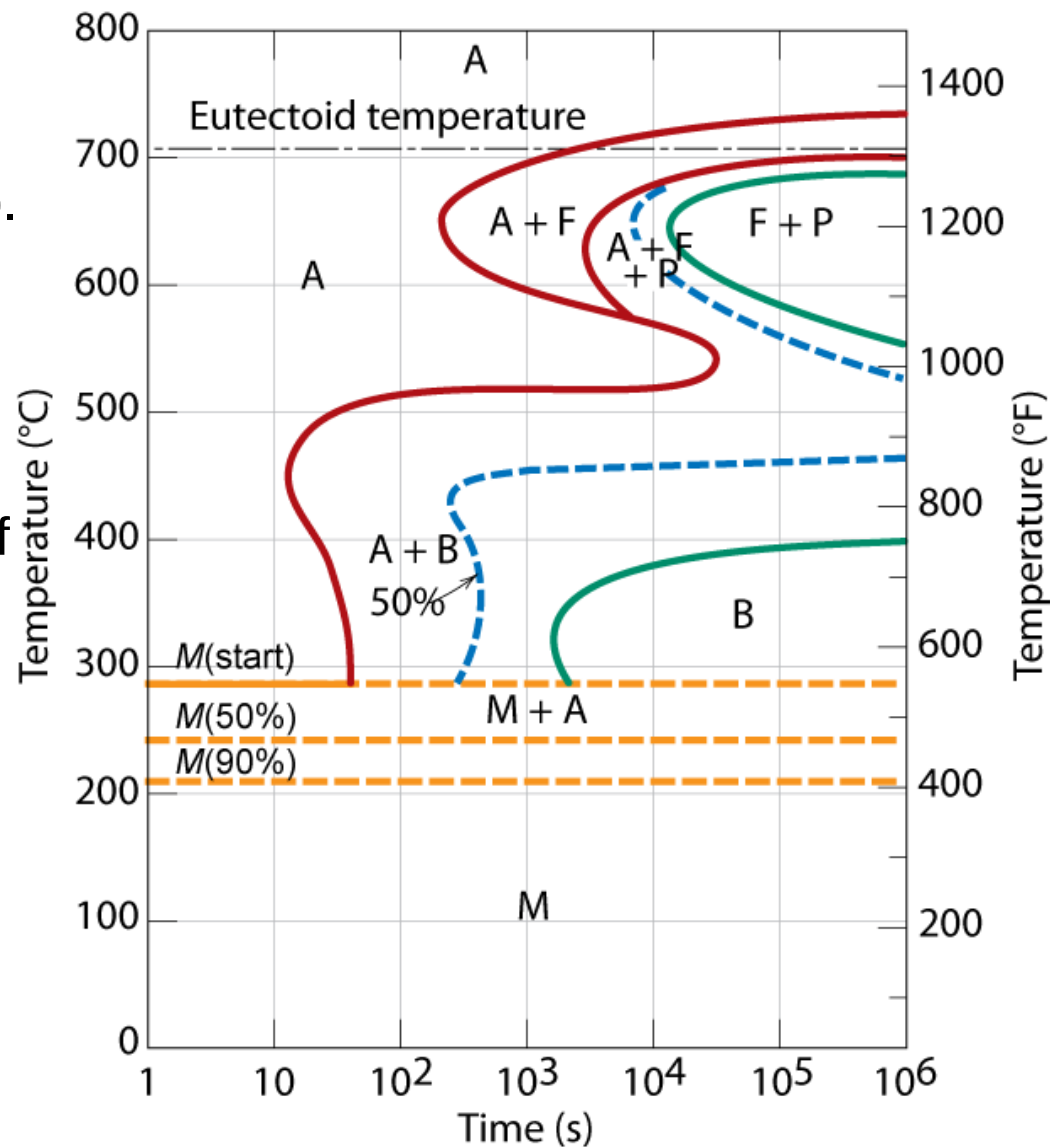
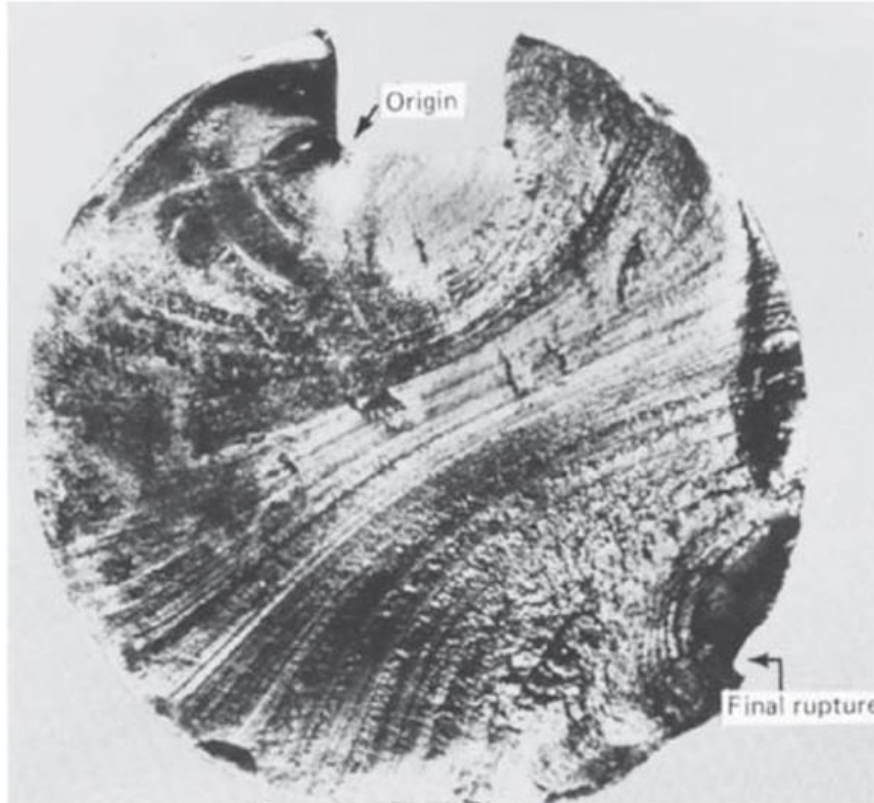


Fig. 10.23, Callister & Rethwisch 9e.
[Adapted from H. Boyer (Editor), *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, 1977.
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Fracture and failure



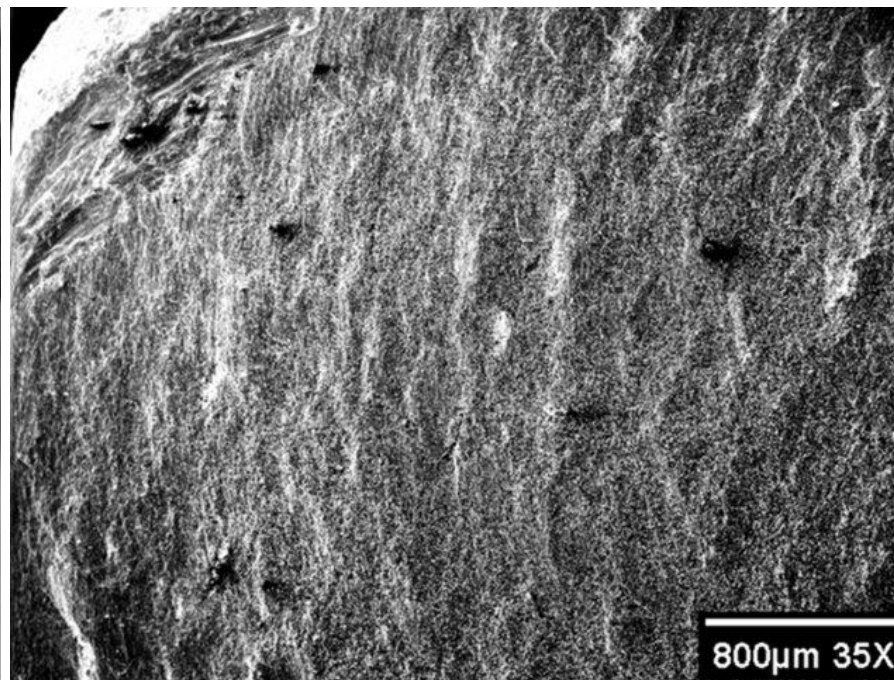
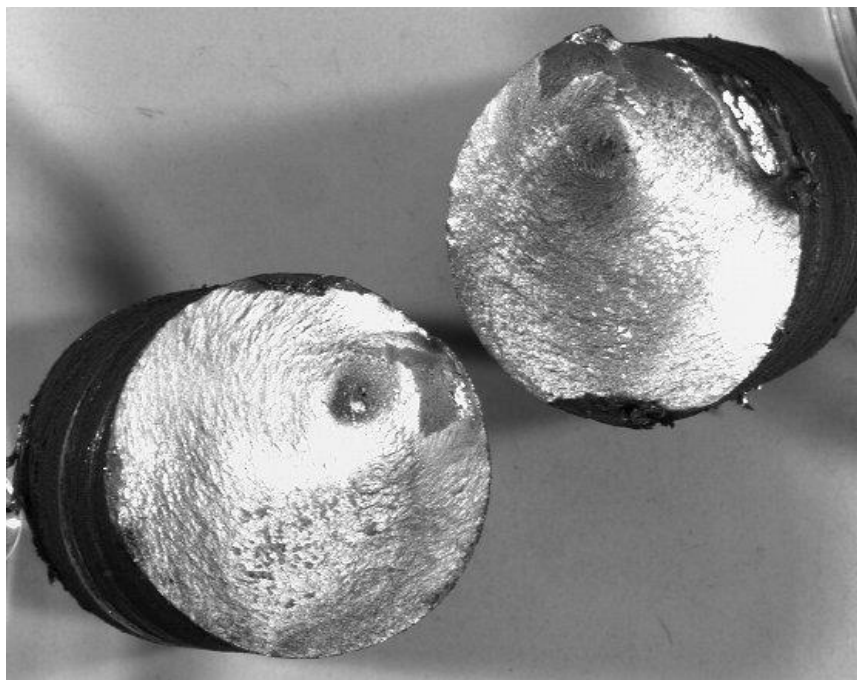
From D. J. Wulpi, Understanding How Components Fail, 1985. Reproduced by permission of ASM International, Materials Park, OH. Wiley, Int.

Fracture surface
of a rotating steel shaft that
experienced fatigue failure.

- Ductile fracture:
plastic
deformation
- Brittle fracture:
rapid crack
propagation
- Fatigue fracture:
due to cyclic
stress

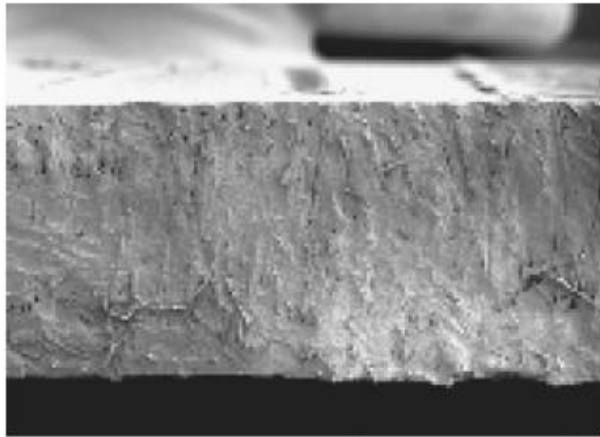


Fracture of A36 Steel





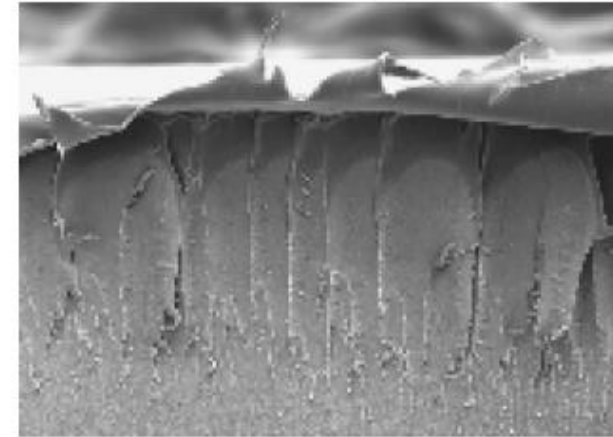
Fracture of PET assemblies



SEM image showing brittle fracture features on a failed jacket cracked surface. 20x



SEM image showing features associated with brittle fracture and slow crack growth within the crack initiation. 100X



SEM image showing features indicative of rapid crack extension within the final fracture zone. 20x

Jansen, J.A. (2002) Conducting a plastic component failure investigation: examples from the appliance industry. International Appliance Technology conference.



Complete the following on-line tutorials

- Crack Theory:

<https://www.unsw.edu.au/science/our-schools/materials/engage-with-us/high-school-students-and-teachers/online-tutorials/crack-theory>

- Mechanical behaviour of polymers

<https://www.unsw.edu.au/science/our-schools/materials/engage-with-us/high-school-students-and-teachers/online-tutorials/polymers>



Review the following processes – Write a summary of the findings

- Read in the book and other provided material the following concepts
 - Fracture and failure
 - Eutectic and eutectoid reactions
 - Hot and cold working
 - General materials properties
 - Mechanical properties
 - Pay attention to units!



Review the following definitions – write a summary of the studied concepts

- Use the Callister book or any other materials science book to review the following definitions:
 - Alloy steel
 - Annealing
 - Cast iron
 - Hardenability
 - Welding
 - Elastomer
 - Glass transition temperature
 - Molding
 - Plasticizer
 - Viscoelasticity